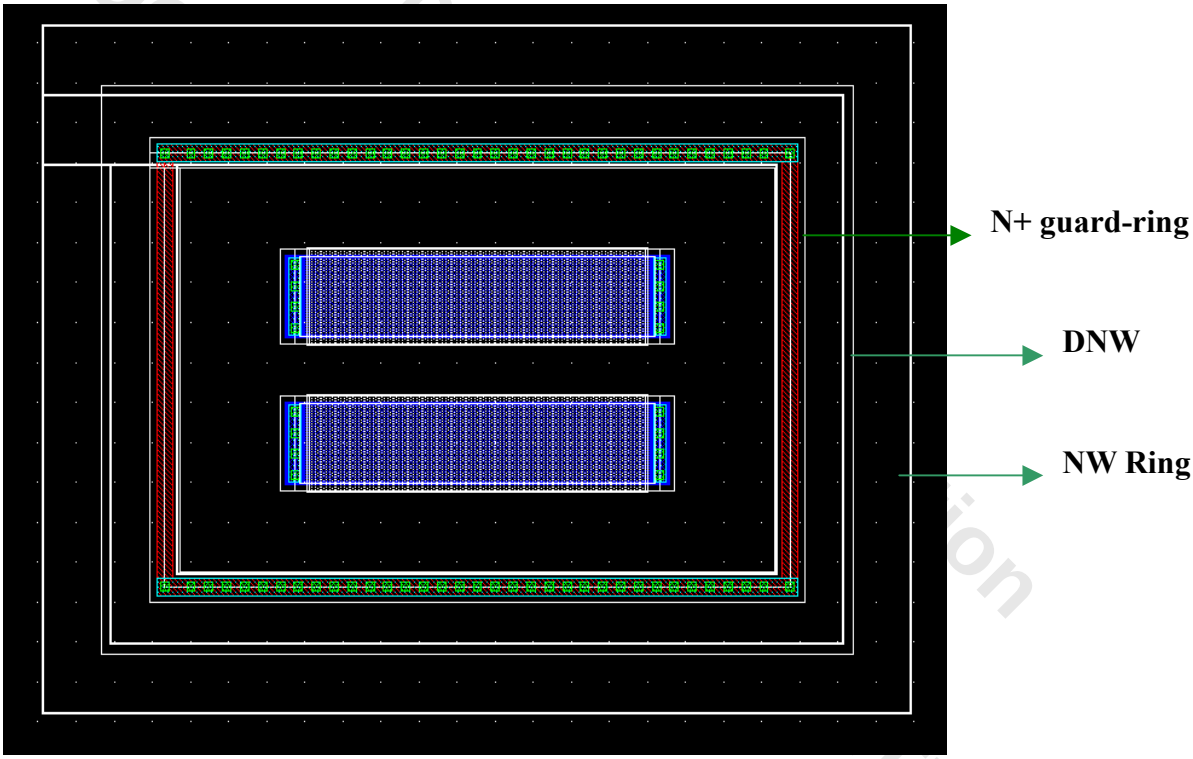
Guard-ring drawn guidelines for RF devices

1. Introduction

This document describes the guard-ring drawn guidelines for RF devices. If users choose to remove the protection guard-ring (P+/NW/DNW) from the original RF devices for devices serial/parallel configuration purpose, they still need to manually add the protection guard-ring after finished the serial/parallel configuration, *such that the LVS command files can correctly recognize the RF devices.*

2. Guard-ring drawn guidelines

A. RF resistors



- N+ guard-ring / DNW / NW-ring have to be manually drew by users
- N+ guard-ring is the 3rd terminal of RF resistors

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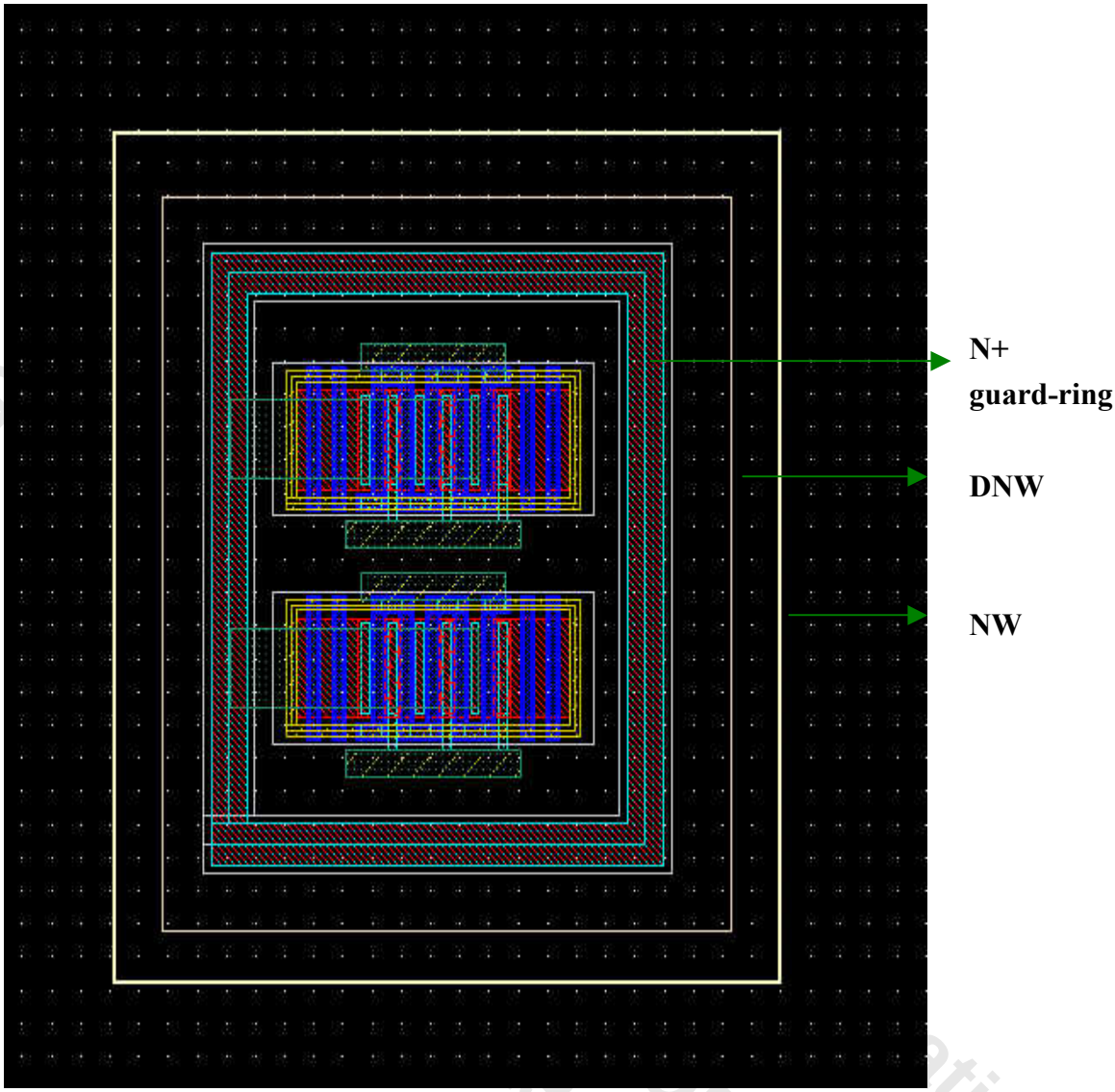
B. RF PMOS



- **N+ guard-ring / DNW / NW have to be manually drew by users**
- **N+ guard-ring is the 'bulk' terminal for RFPMOS**

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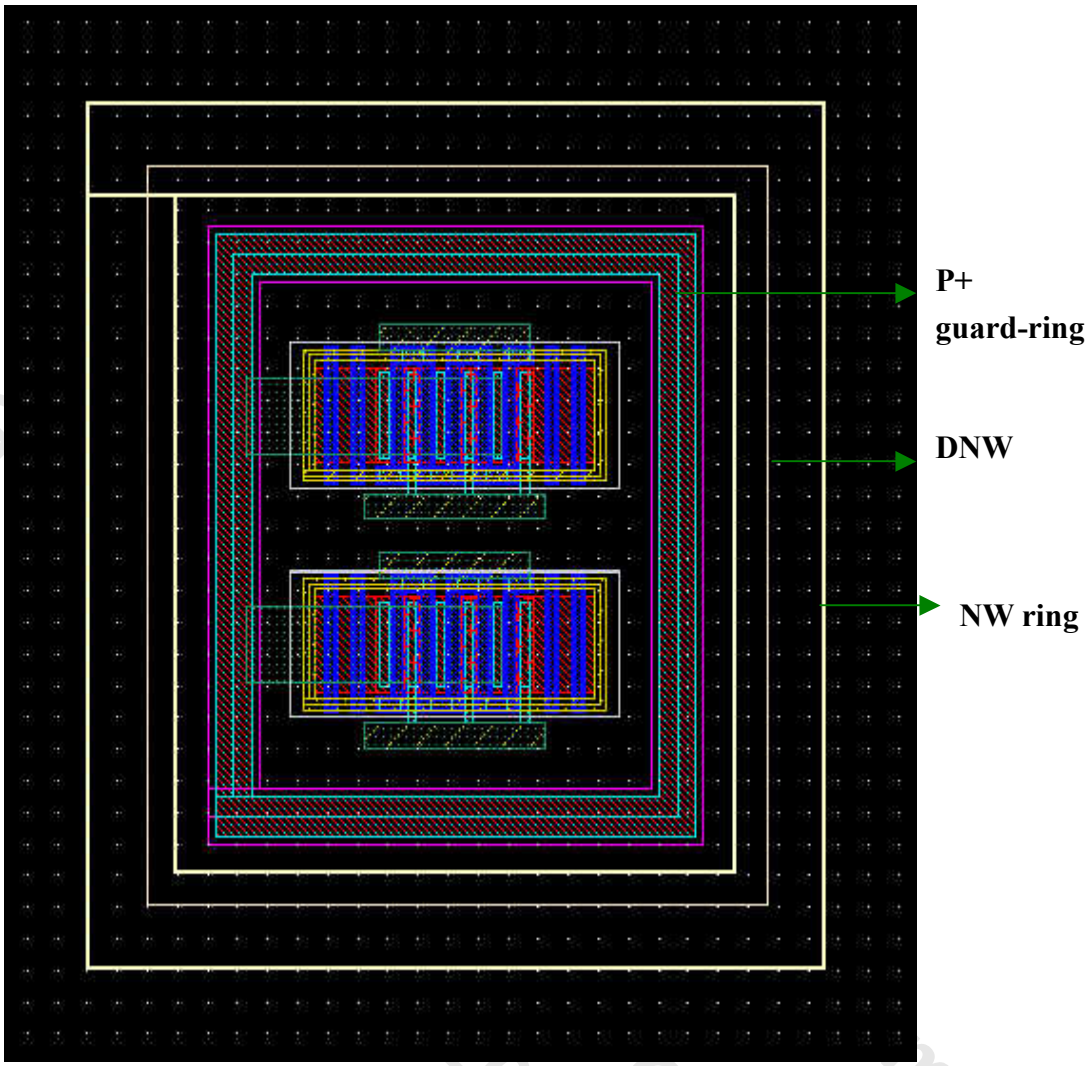
C. RF NMOS



- P+ guard-ring / DNW / NW-ring have to be manually drew by users
- P+ guard-ring is the 'bulk' terminal for RFNMOS

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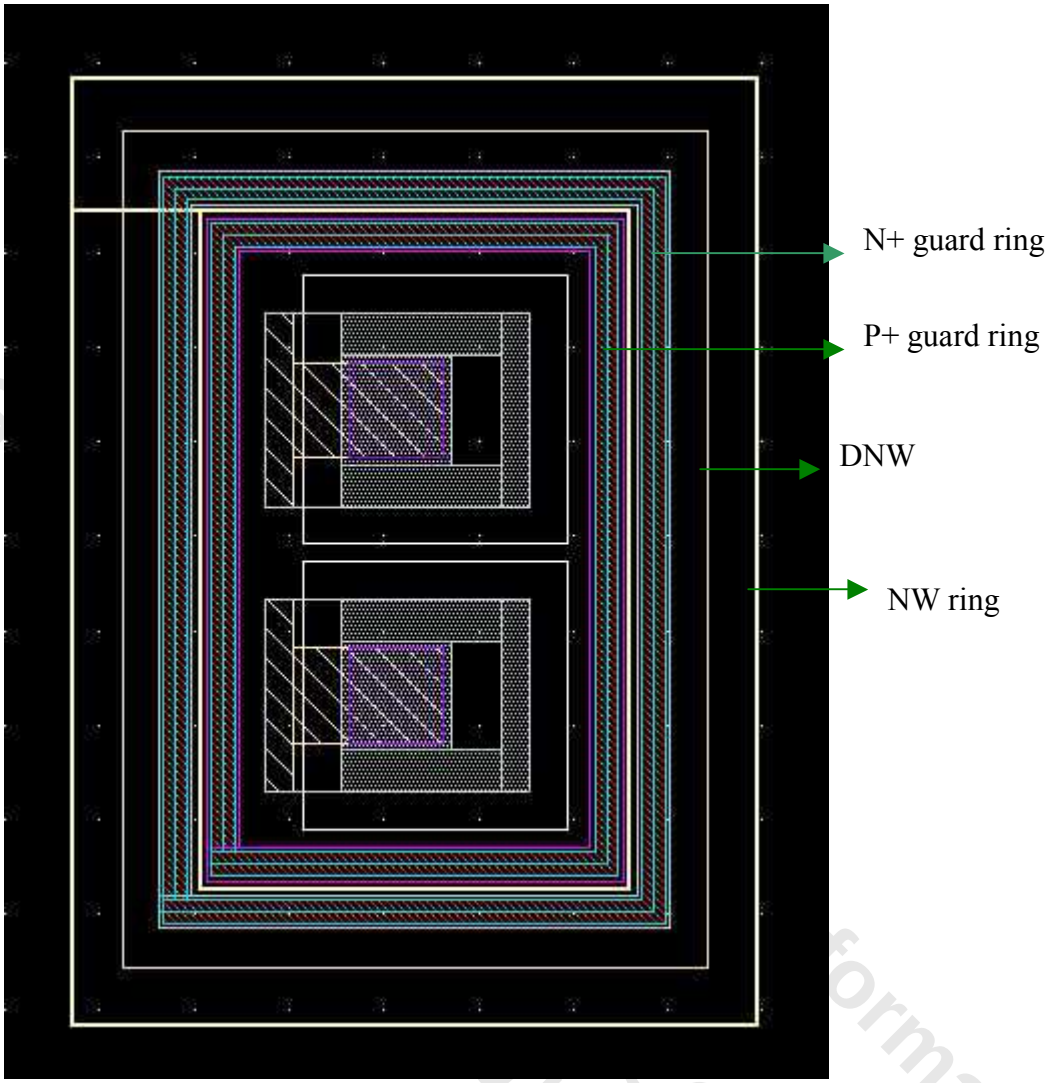
D. MIMCAP_WOS (mimcap w/o shield)



- **P+ guard-ring / N+ guard-ring / DNW / NW-ring have to be manually drew by users**
- **P+ guard-ring and N+ guard-ring must tie together. (LVS purpose)**
- **N+ guard-ring is the 'bulk' terminal of mimcap_wos device.**

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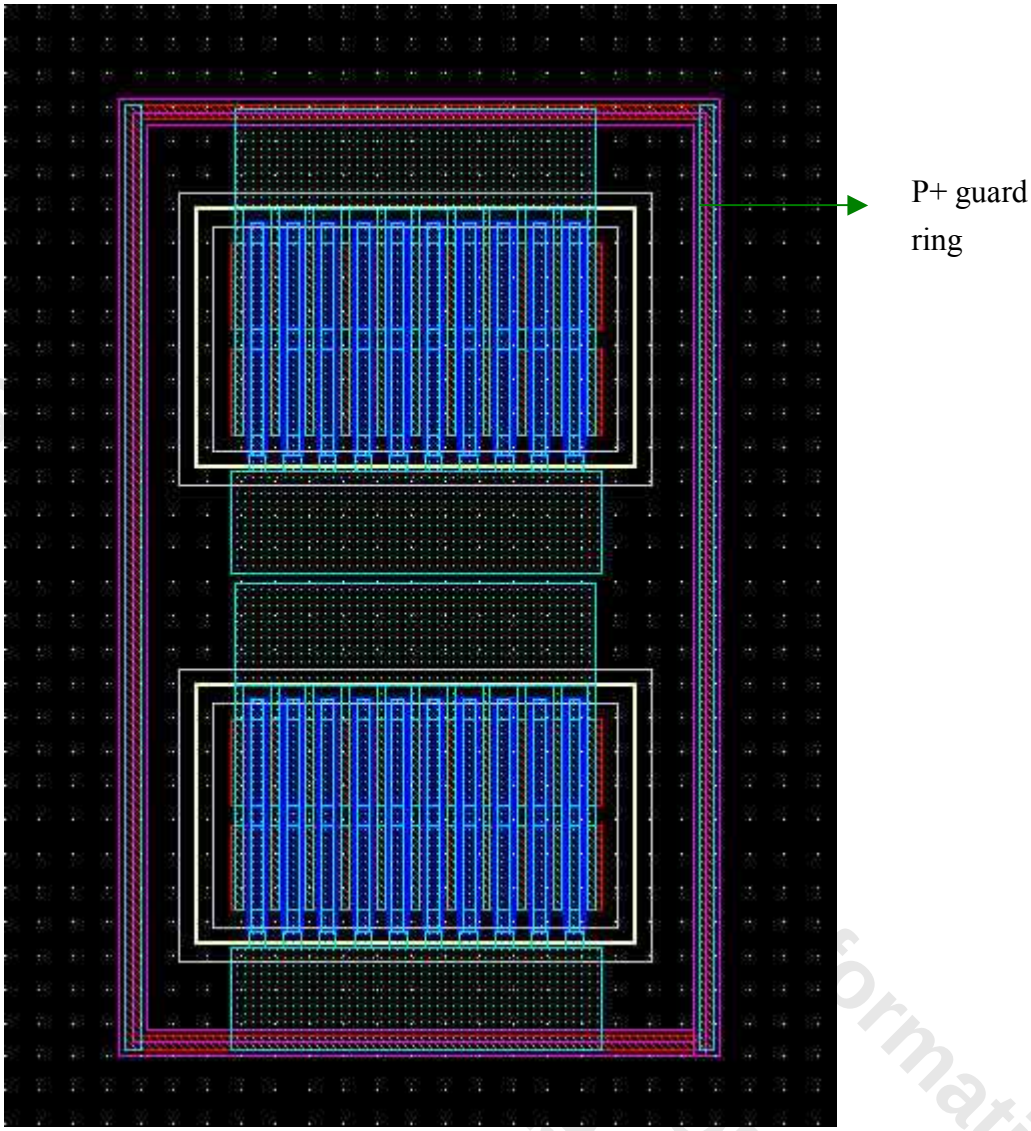
E. Varactor (junction/mos varactors)



- **P+ guard-ring has to be manually drew by users**
- **P+ guard-ring is the 3rd terminal of varactor**

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